The Minimum Tileset Problem for $2 \times n$ Rectangles

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Abstract

We study the number of tile types needed to uniquely self-assemble a $2 \times n$ rectangle in the abstract Tile Assembly Model (aTAM). At temperature 2, the answer is $\Theta(\sqrt{n})$ for every seed position. The lower bound is known, and the interior-seed upper bound is a sketch. We add a comb with $n + 1$ types for every seed cell and a two-row counter with about $3.46\sqrt{n}$ types, hand-traced but not machine-simulated. For unary input, Mahaney's theorem shows that the problem is not NP-complete unless P = NP. We also give a $2^{O(\sqrt{n} \log n)}$ exact algorithm and an about 8-approximation. The exact values for the interior seed and for temperature 1 stay open.

1 Introduction and Model

The Minimum Tileset Problem asks for the fewest tile types that uniquely self-assemble a given shape. The general problem is NP-complete, and it is polynomial for squares at temperature 2 and at every temperature.[1] The square result is from [3]. The strict (non-directed) variant is Σ_2^P-complete [2]. We ask for the complexity of the directed problem on $2 \times n$ rectangles. Our answer is a tight $\Theta(\sqrt{n})$ range on the number of types, plus two complexity statements.

Model. A tile type is a unit square with a glue label on each of its four sides (N, E, S, W), and a glue function gives each label a strength. Tiles never rotate and are placed by translation only. At temperature τ, a tile attaches when its matching sides have total strength at least τ. One seed tile sits at a fixed cell, and growth starts there. We call a system *directed* if it has exactly one terminal assembly and that assembly has the target shape. Let t be the least number of tile types of a directed system for $2 \times n$ at a stated τ and seed cell. We state τ for every result. Where input size matters, n is given in unary or as a sorted list of cells, and s is in unary or fixed width. The number k may be in binary, since the reduction in Section 3 rewrites it.

2 Results

Table 1 lists every bound with its true status. The two-row answer is $\Theta(\sqrt{n})$ at $\tau = 2$ for each seed position.

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[1] The NP-completeness is due to Adleman et al. (STOC 2002), which we read only in abstract form.

Setting	Lower	Upper	Status
2×1, any τ	2	2	proved (Thm. 1)
$2 \times n$, any τ, any seed cell	$\sqrt{n}/2$	$n+1$ (comb)	lower: [1]. Upper: proved (Thm. 2)
$\tau = 2$, corner seed	$\sqrt{n}/2$	$4\lceil\sqrt{n}\rceil + 2$	proved [1]
$\tau = 2$, corner seed, $n = 1 + mD$	$\sqrt{n}/2$	$m + 3D + 2 \approx 3.46\sqrt{n}$	construction, hand-traced, not machine-simulated
$\tau = 2$, interior seed	$\sqrt{n}/2$	$4\sqrt{2}\,\sqrt{n} + O(1)$	upper bound is a sketch
$\tau = 1$, no mismatches	$n+1$	$n+1$	exact, from a source we read only in part
$\tau = 1$, general	$\Omega(\sqrt{n})$	$n+1$	exact value open

Table 1: Bounds on the number of tile types t for a directed $2 \times n$ rectangle. Why: it separates what is proved from what is only sketched or hand-traced.

2.1 Small cases and the comb

Theorem 1. *For every $\tau \geq 1$, the 2×1 rectangle needs exactly 2 tile types.*

Proof. Take one type with glues N and S. The second tile binds to the seed through one side, so that glue has strength at least τ. Then N and S carry the same label, and a third copy attaches above the second. Growth never stops. Two types suffice: A has north glue a of strength τ, and B has south glue a. All other sides are null. □

Theorem 2. *For every $\tau \geq 1$, every n and every seed cell, a $2 \times n$ rectangle is directed by $n+1$ tile types.*

Proof. Let the bottom row use types $B_1, \ldots, B_n$. Type B_i has east glue e_i for $i < n$, west glue e_{i-1} for $i > 1$, and north glue u. One top type T has south glue u. Every other side is null, and every glue has strength τ. Take the seed B_s. East of it, only B_{s+1} has west glue e_s. West of it, only B_{s-1} has east glue e_{s-1}. By induction, each bottom cell has one candidate. Each top cell offers only T, and the null east and west sides of T stop growth. So the terminal assembly is unique with shape $2 \times n$. A top-row seed breaks this set, because the n bottom types compete under the seed T. Use the vertically flipped comb: top types $T_1, \ldots, T_n$ with the chain glues e_i and south glue u, and one bottom type B with north glue u. The same induction applies. □

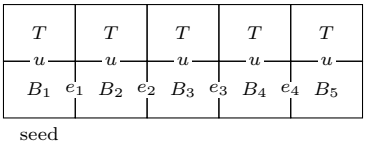


Figure 1: The comb for $n = 5$ with the seed at B_1. Why: the five bottom tiles are chained by e_i, and all five share one top type T.

2.2 Lower bound and corner upper bound

Theorem 3 ([1], Thm. 3.1)**.** *At every τ and wherever the seed lies, a directed $2 \times n$ system has $t \geq \sqrt{n}/2$.*

We do not re-prove it. The proof finds three identically placed columns with the same attachment order, two on one side of the seed, and pumps the stretch between them. For a corner seed at $\tau = 2$, a base-m two-digit counter with $m = \lceil\sqrt{n}\rceil$ gives $t \leq 4\lceil\sqrt{n}\rceil + 2$ [1]. For $n = 100$, this reads $5 \leq t \leq 42$. Thus the answer is $\Theta(\sqrt{n})$ at $\tau = 2$ for a corner seed, and the constants differ by a factor of about 8.

2.3 A leaner two-row counter

Construction (hand-traced, not machine-simulated). Let $n = 1 + mD$ with $m \geq 2$ and $D \geq 1$. The seed Z is the bottom tile of the east column, and growth runs west in D blocks of m columns. The bottom row counts the position in a block with a chain $P_1, \ldots, P_m$. Here P_j has west glue p_j and P_{j+1} has east glue p_j, both of strength 2. Strength 2 is needed, since strength 1 would deadlock P_{j+1} against the tile above it. The top row holds a digit d that counts blocks down from $D - 1$ to 0. Each d has its own strength-1 label δ_d and a strength-2 label r_d. A shared digit label would let two digit tiles fit one cell. The block cycle has five steps.

1. Start tile S_d attaches on r_d alone.
2. P_1 attaches under it on $q + s$ with strengths $1 + 1$.
3. $P_2, \ldots, P_m$ follow on the strength-2 chain glues.
4. Middle tile M_d fills each top cell on $\delta_d + g$.
5. Decrement tile X_d closes the block on $\delta_d + b$ and emits r_{d-1}.

After X_0, no start tile exists, and P_1 sees only q of strength 1, so growth stops. The tile types are m of type P, D each of S, M, X, and two seed-column tiles Z and Y, so $m + 3D + 2$ in all. Setting $m = 3D$ gives $2\sqrt{3n} \approx 3.46\sqrt{n}$ types, against $4\sqrt{n} + 2$ in [1]. We traced the construction by hand for $(m, D) = (3, 2)$ and $(12, 4)$. For $(12, 4)$, $n = 49$ and the count is 26, against 50 for the comb. For $D = 58$ and $m = 173$, $n = 10{,}035$ and the count is 349. The construction covers only $n = 1 + mD$, and other lengths need a partial first block that we did not work out. We claim it as a construction, not as a theorem.

2.4 Interior seed and temperature 1

Proposition 1 (sketch)**.** *At $\tau = 2$ and seed column s, $t \leq 4\lceil\sqrt{a}\rceil + 4\lceil\sqrt{b}\rceil + O(1) \leq 4\sqrt{2}\,\sqrt{n} + O(1)$, where $a = s - 1$ and $b = n - s$.*

Sketch. Grow one counter west and its mirror image east, sharing the seed column and using disjoint glue labels. The step to the second inequality is $\sqrt{a} + \sqrt{b} \leq \sqrt{2(a + b)}$. The sketch relies on the claim of [1] (p. 8) that the counter handles every arm length from 1 to m^2, which we did not re-check. For a top-row seed cell, flip the construction vertically. With Theorem 3, this gives $\Theta(\sqrt{n})$ at $\tau = 2$ for every seed position.

At $\tau = 1$, a result of Maňuch, Stacho and Stoll states that a directed system without binding-domain mismatches needs at least the Manhattan diameter plus 1 types. The diameter of $2 \times n$ is n, so the exact value is $n + 1$ under that assumption, and the comb is mismatch-free. This part rests on a source we read only in part. Without the assumption, the known bounds are $\Omega(\sqrt{n})$ and $n + 1$ [4]. Theorem 3 gives the same $\Omega(\sqrt{n})$ at every τ.

3 Complexity

Theorem 4. *Fix τ, and give n and s in unary or fixed width, or the shape as a sorted cell list. The number k may be in binary. Then deciding $t \leq k$ for $2 \times n$ is not NP-complete unless $P = NP$.*

Proof. First map any encoding of the shape to the canonical form $\langle 1^n, s, k\rangle$ in polynomial time. Send every instance with $k \geq n + 1$ to one fixed yes-instance, since the comb gives $t \leq n + 1$. Send every non-rectangle to one fixed no-instance. Now each input length has only polynomially many strings, since $s \leq 2n$ and $k \leq n + 1$. So the canonical language is sparse. If the original problem were NP-hard, the canonical language would be NP-hard under many-one reductions. Then Mahaney's theorem gives P = NP. □

The argument fails only for n in binary, since then the canonical language is not sparse. A binary k in the input is fine, because the reduction maps $k \geq n + 1$ to a fixed yes-instance and rewrites k in unary or fixed width. NP membership needs a fixed τ, since a certificate is a tile set of at most $n + 1$ types.

Proposition 2 (algorithm and approximation)**.** *At fixed $\tau = 2$ with unary or cell-list input, an exact algorithm runs in time $2^{O(\sqrt{n}\log n)}$.*

Sketch. The optimum is at most $4\lceil\sqrt{n}\rceil + 2$ for a corner or free seed. A system with t types has $4t$ sides, each with one of $O(t)$ labels. That gives $t^{O(t)} = 2^{O(\sqrt{n}\log n)}$ systems, and each is checked in polynomial time by uniqueness verification. For a seed cell fixed by the input, search up to the bound of Proposition 1. For binary n this is not subexponential, and at $\tau = 1$ the only bound is $n+1$, so the same search costs $2^{O(n\log n)}$. For a free seed, output the corner counter of [1] against the bound $\sqrt{n}/2$. The ratio tends to about 8 and stays below about 11 for small n. The counter of Section 2 gives about 7 when $n = 1 + mD$. For a seed cell fixed by the input, Proposition 1 gives about $8\sqrt{2} \approx 11.3$.

4 Open Problems

1. The exact corner-seed formula that the problem statement mentions. We did not find it, and our range is between $\sqrt{n}/2$ and $4\lceil\sqrt{n}\rceil + 2$.
2. The exact interior-seed value, and whether two arms can share tile types so that an interior seed beats a corner seed.
3. The exact $\tau = 1$ value without the no-mismatch assumption, between $\Omega(\sqrt{n})$ and $n + 1$.

Exact small-n values need an exhaustive search that we did not run.

AI Usage Statement

An AI assistant found and read sources, drafted the text, and hand-checked the constructions and statements above. It ran no simulation, so the counter of Section 2 is hand-traced only.

References

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